



GREATER NOIDA INSTITUTE OF TECHNOLOGY



GESTURE-BASED VIRTUAL MOUSE

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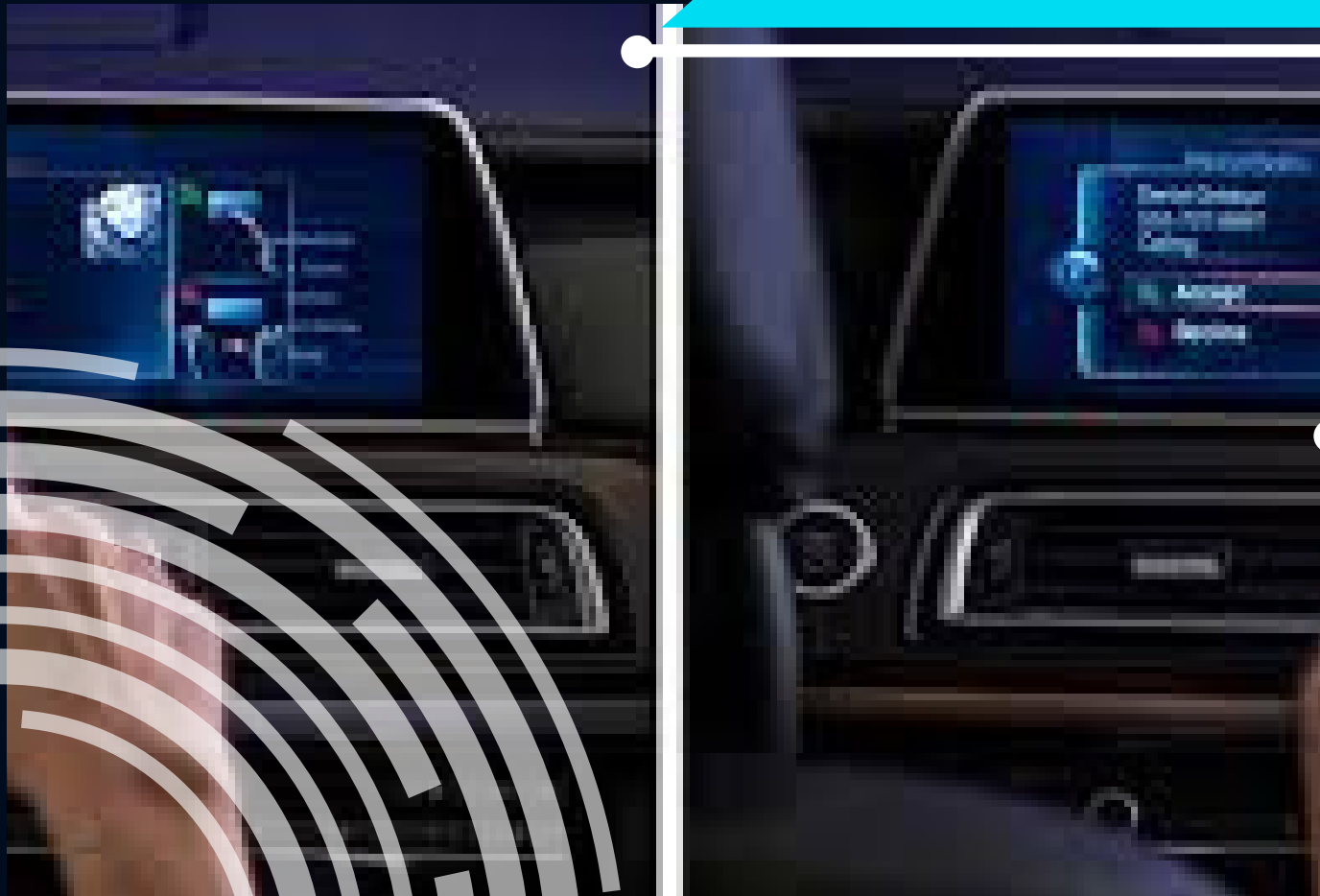
HOD, Computer Science & Engineering (AI)

Introduction

what is Hand Gesture recognition

A technology that interprets human hand movements using computer vision, sensors, and AI

Use an icon or animation of a hand or camera.



Why it's needed:

- Hands-free device control
- Easy Human-Computer Interaction
- Safe for healthcare & industries
- Helpful for disabled users
- Enhances gaming, AR/VR & robotics
- Smart & touchless future



Vision

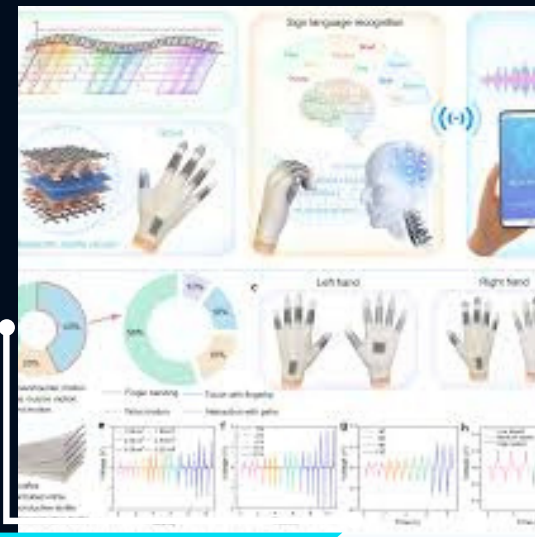
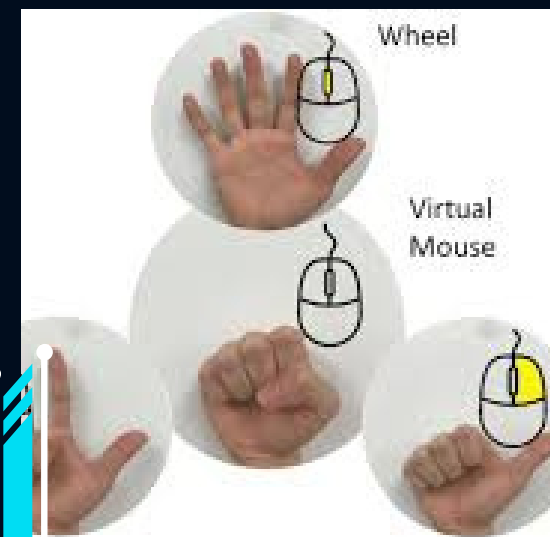
To Create a World Where humans can interact with technology naturally and effortlessly, using just their gesture

Mission

TO make technology accessible for everyone ,including people with disabilities.

• Tools and Technologies Used:

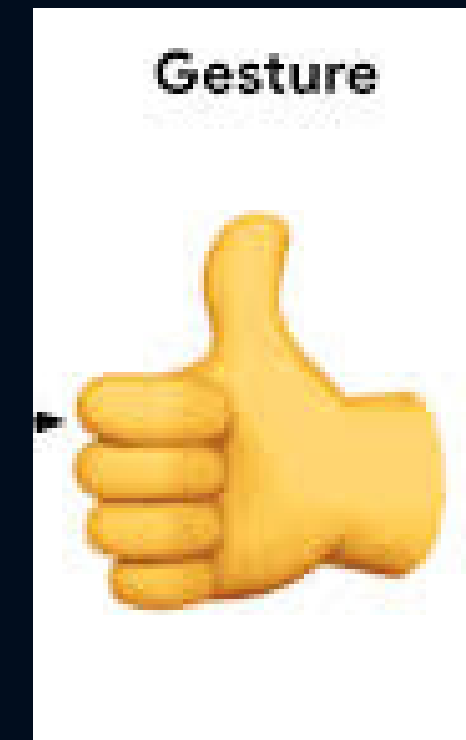
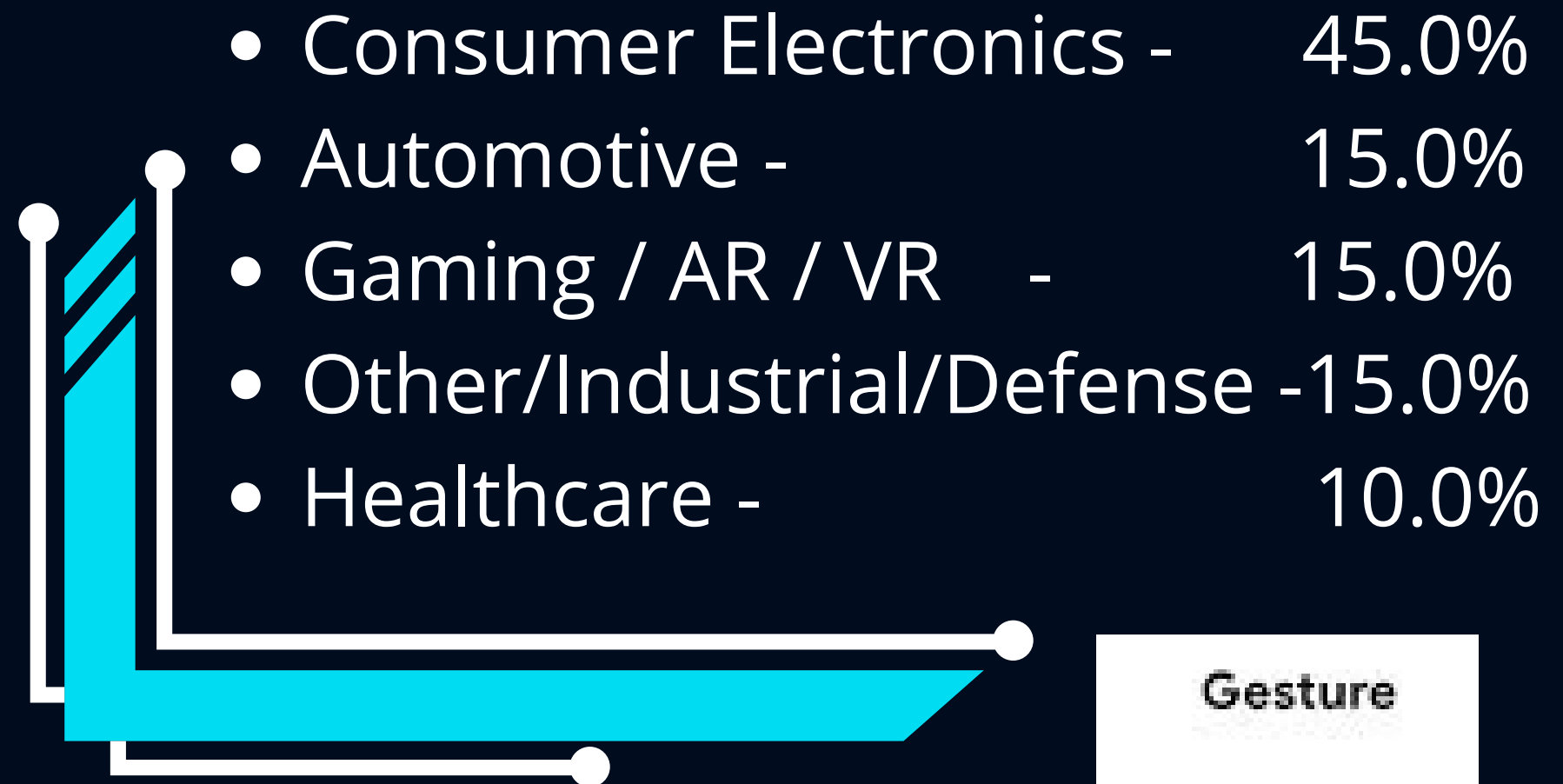
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- Component Technology
- Programming Language: : Python
- Computer Vision: : OpenCV
- Hand Tracking: : Mediapipe
- IDE/IDLE : VS Code



Applications

- Gaming (gesture-controlled games .)
- Robotics (robot arm control)
- Healthcare (sign language...)
- Smart Devices (IoT, AR/VR)
- Number Recognition (like-using real time system HMM)
- 3-D Modeling,
- Television Control...
-

Field distribution



Challenges:

Environmental Factors: Accuracy can suffer due to lighting, background complexity, and motion, making robust recognition in unpredictable real-world settings difficult.

Dynamic Gestures: Distinguishing and tracking dynamic and complex gestures, especially in continuous motion, remains challenging for current algorithms.

Hardware Requirements: Most systems require specialized cameras or sensors, and high computational power for real-time processing.



Future Scope:

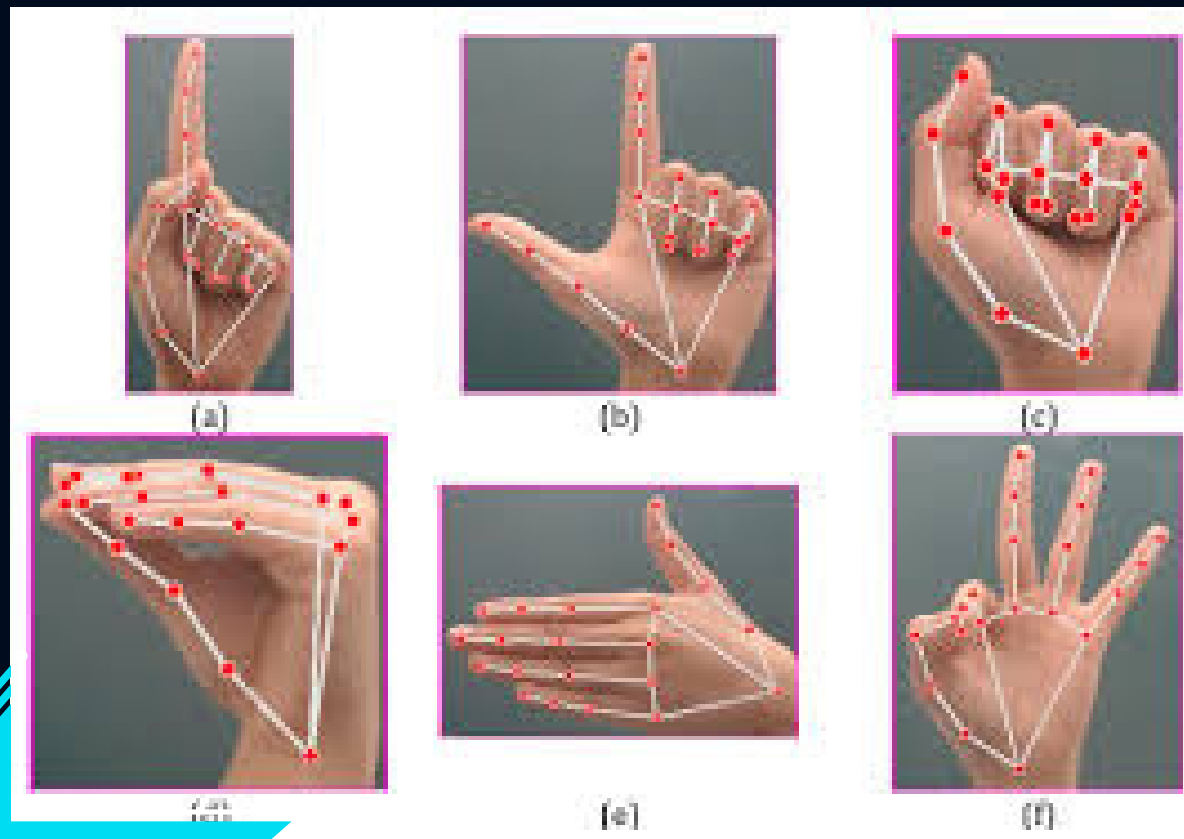
Healthcare Applications: Gesture-based systems can aid remote patient monitoring, rehabilitation, and assistive technology for the differently-abled.

Smart Environments: Integration into IoT and smart home devices will allow touchless control, enhancing convenience and hygiene

- More accurate AI models
- Integration with AR/VR
- Touchless UI for smart cities
- Healthcare innovations

WITH AN PERFECT EXAMPLE...--->

Real life example



Conclusion:

Gestures are the future of seamless human-computer interaction.

Synthetic data helps train face recognition systems safely and easily. It gives privacy, variety, and better results, but we still need to fix issues like fairness and misuse. In the future, real and fake data together will make AI more accurate and safe."



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THANK YOU

RMR TEAM